Network Management and Administration (IE2110)
2nd Year, 2nd Semester

Assignment 01

Zimbra

Submitted to

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Bachelor of Science Special Honors Degree in Information Technology

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1. *Initial Setup - Part A*

1.1 VM Creation

In this step, four virtual machines were created using VirtualBox to simulate a network environment. Two virtual machines was configured as servers using the Rocky Linux and Ubuntu operating systems, while the other two virtual machines were configured as client systems using Ubuntu.

Appropriate resources such as CPU, memory, and storage were allocated to each virtual machine to ensure smooth operation of the required services.

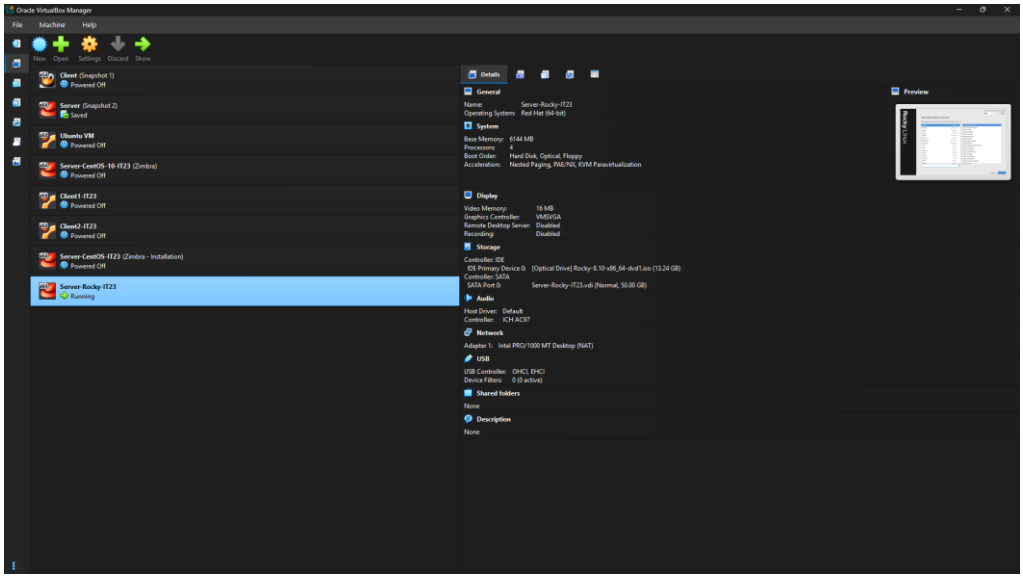

Figure 1.1: Server Rocky Linux

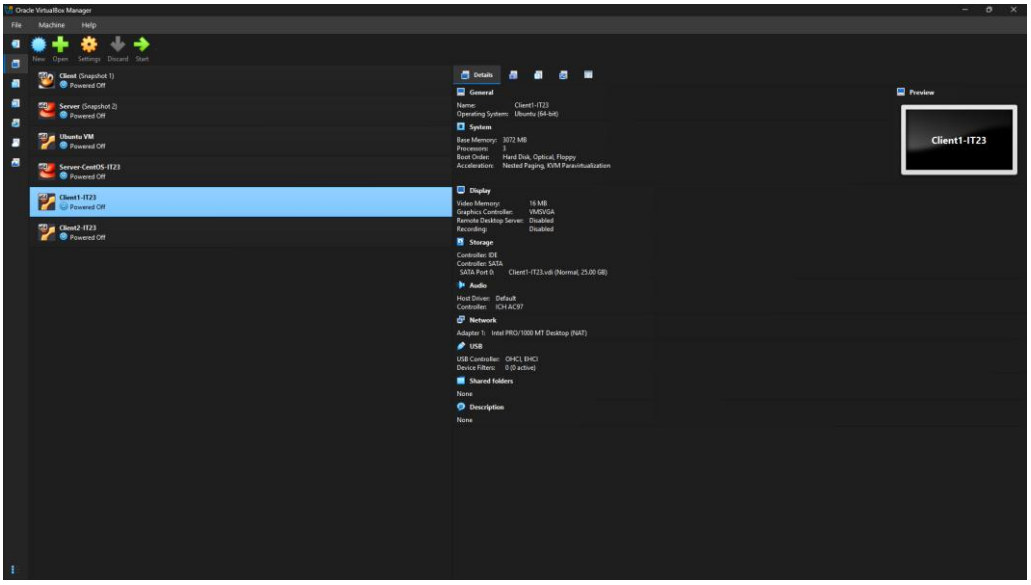

Figure 1.2: Ubuntu Client 01

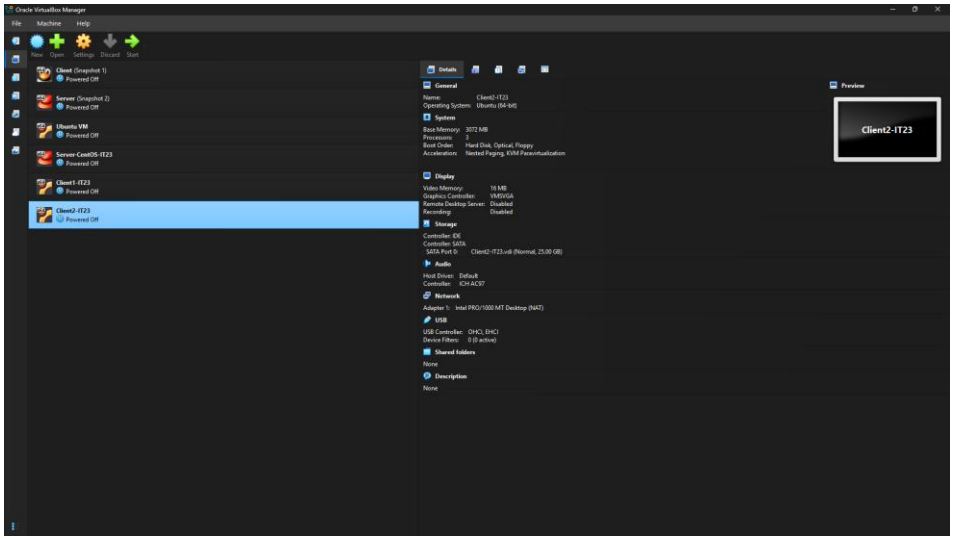

Figure 1.3: Ubuntu Client 02

- These screenshots show the creation of the Virtual machines in VirtualBox. The machine was created with the required operating system and hardware settings to support the network services.

1.2 Operating System Installation

After creating the virtual machines, the respective operating systems were installed on each machine. Rocky Linux and Ubuntu was installed on the server machines, and Ubuntu was installed on both client machines.

The installation process included configuring system settings such as language, storage, and user accounts. All installations were completed successfully, and each virtual machine was able to boot into its respective operating system.

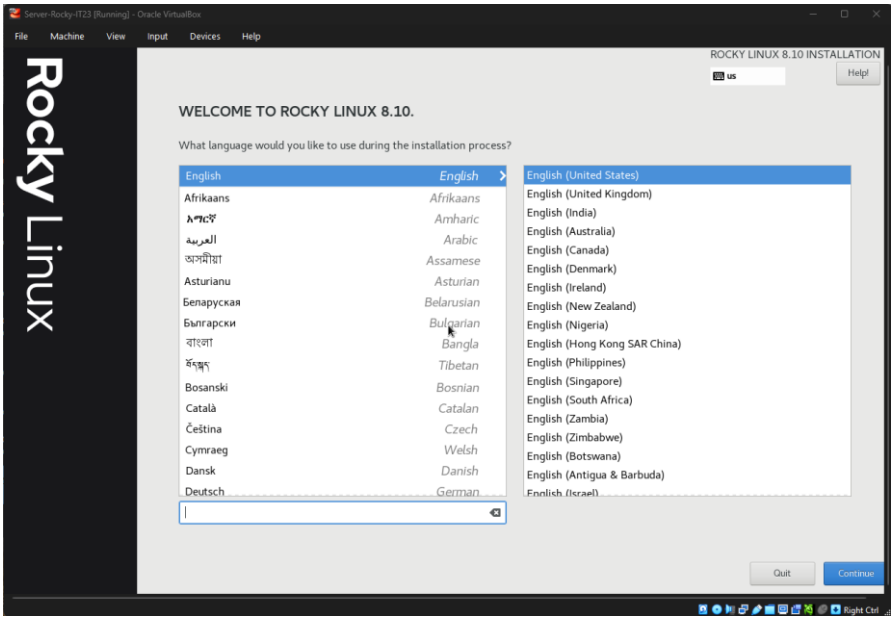

Figure 1.4: Server Rocky Linux OS Installation

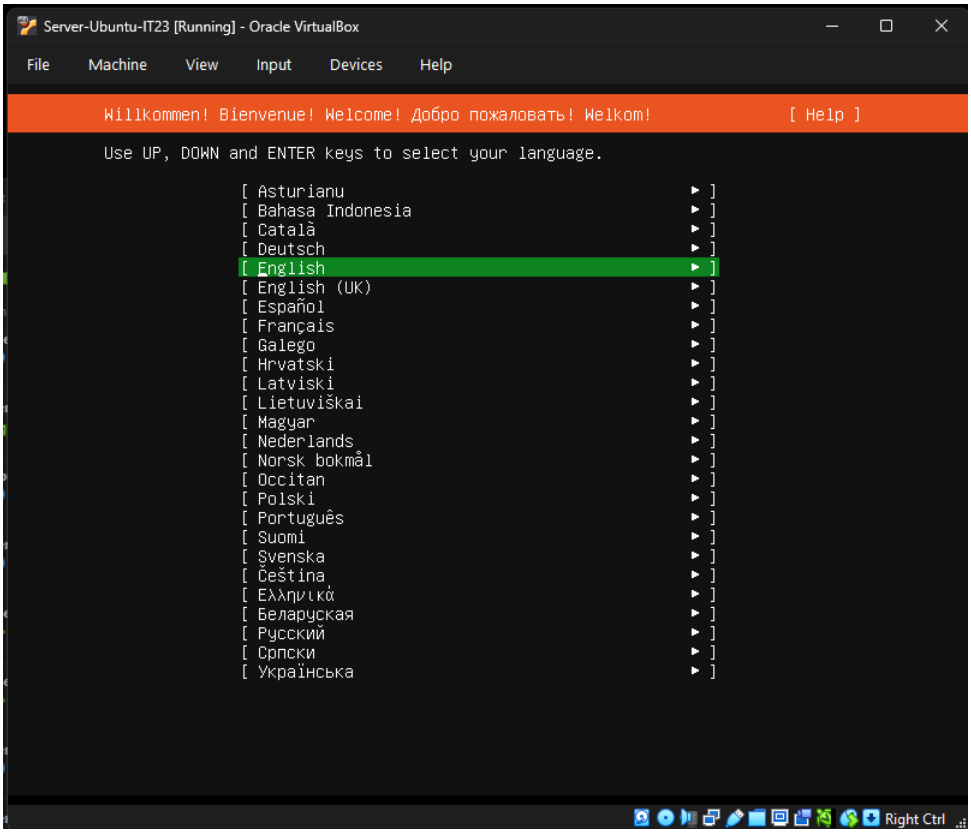

Figure 1.5: Server Ubuntu OS Installation

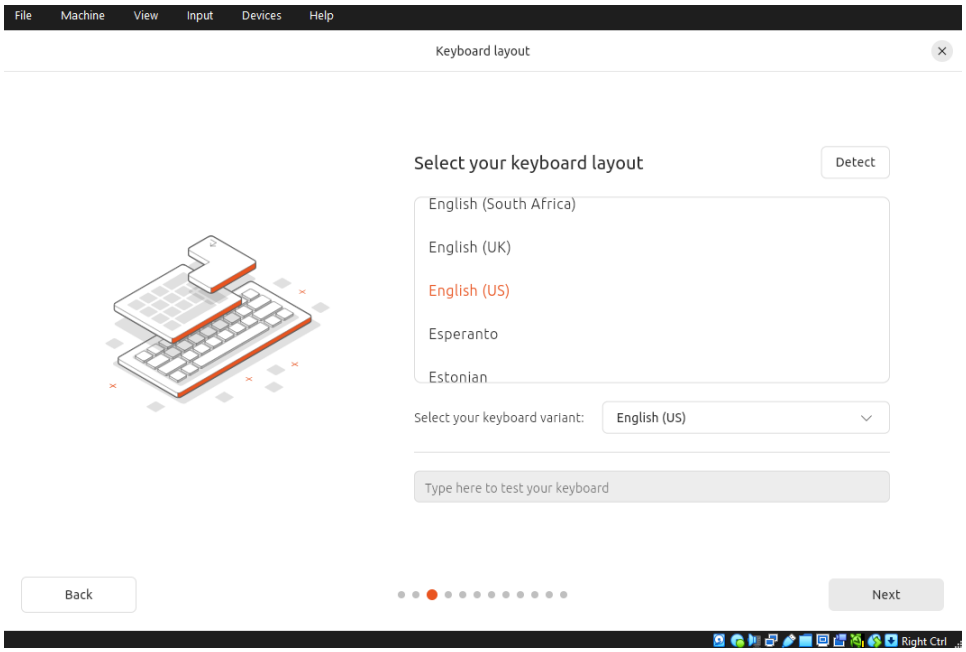

Figure 1.6: Client OS Installation

- These screenshots show the installation of the operating system on the virtual machine. The machine is ready for network and service configuration.

1.3 Network Configuration

In this step, network configurations were applied to ensure that all virtual machines were connected within the same network. Each machine was assigned to the same subnet (192.168.23.0/24) to enable communication between them.

Network connectivity was verified using the `ping` command. Successful responses confirmed that all machines could communicate with each other, indicating that the network setup was correctly configured.

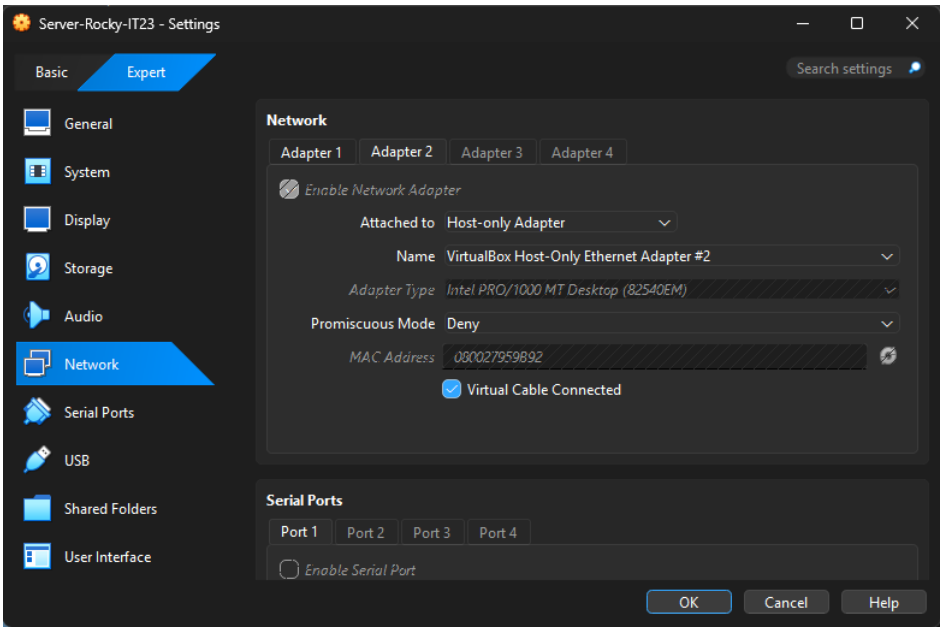

Figure 1.7: Server (Rocky) - Configure Host Only Network

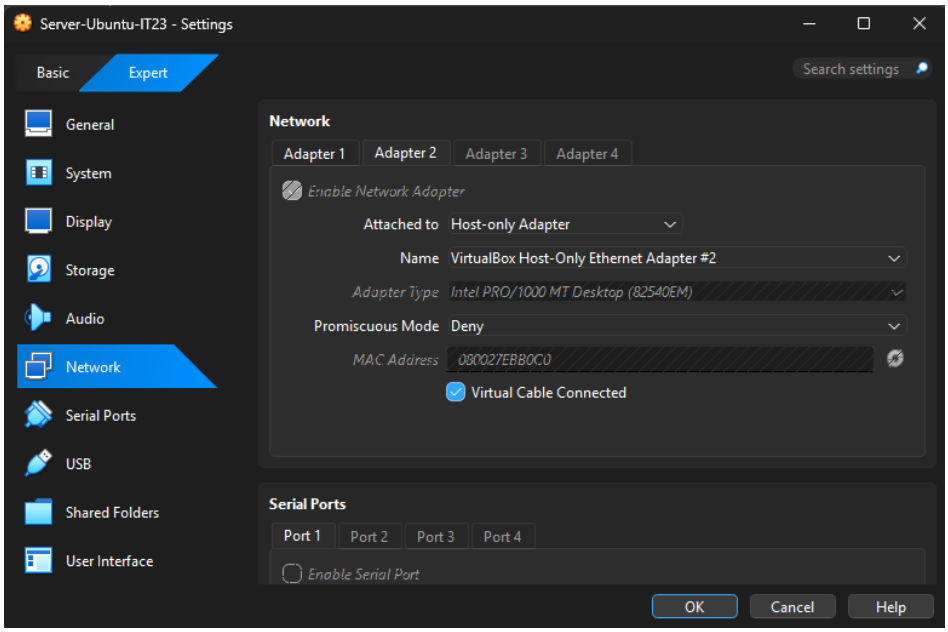

Figure 1.8: Server (Ubuntu) – Configure Only Network

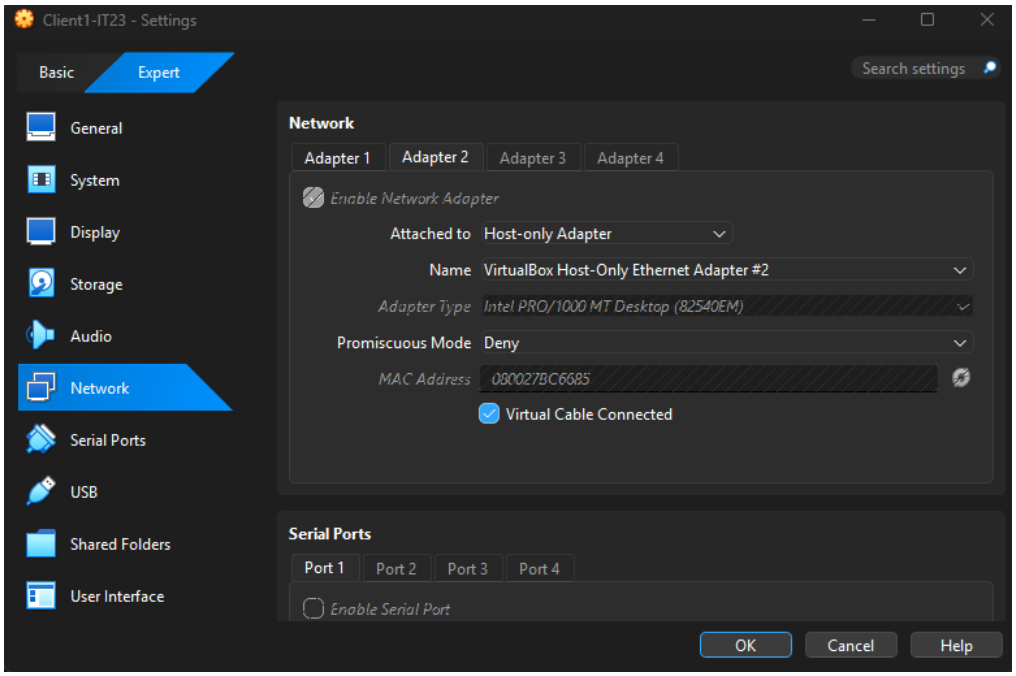

Figure 1.9: Client 01 - Configure Host Only Network

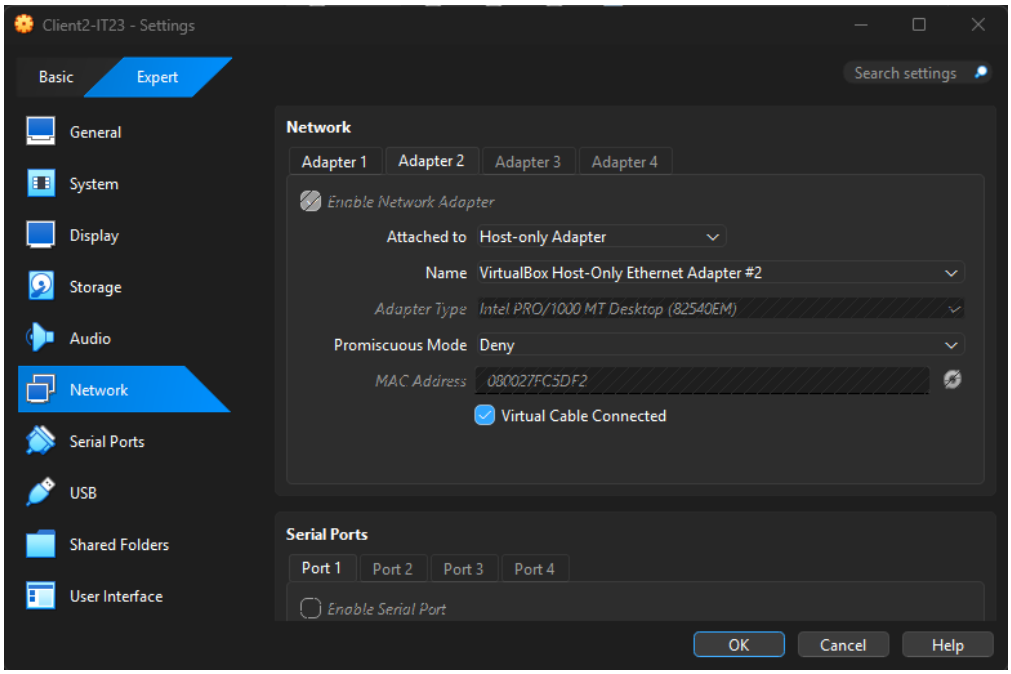

Figure 1.10: Client 02 - Configure Host Only Network

- These screenshots show the network configuration of the virtual machines. The adapter was configured so that all machines can communicate within the same virtual network.

```
[root@localhost ~]# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1
000
    link/ether 08:00:27:6f:5e:60 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global noprefixroute dynamic enp0s3
       valid_lft 85686sec preferred_lft 85686sec
    inet6 fd17:625c:f037:2:46c9:574d:a4c1:335e/64 scope global noprefixroute dynamic
       valid_lft 86114sec preferred_lft 14114sec
    inet6 fe80::95ec:40a:2e9d:d246/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1
000
    link/ether 08:00:27:f3:c7:95 brd ff:ff:ff:ff:ff:ff
    inet 192.168.23.10/24 brd 192.168.23.255 scope global noprefixroute enp0s8
       valid_lft forever preferred_lft forever
    inet6 fe80::9f7e:2399:52b:7c99/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
[root@localhost ~]# _
```

Figure 1.11: Server IP Details (Rocky)

```
it23@ubuntu-it23:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 100
0
    link/ether 08:00:27:92:e7:3a brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
       valid_lft 85700sec preferred_lft 85700sec
    inet6 fd17:625c:f037:2:a00:27ff:fe92:e73a/64 scope global dynamic mngtmpaddr noprefixroute
       valid_lft 86312sec preferred_lft 14312sec
    inet6 fe80::a00:27ff:fe92:e73a/64 scope link
       valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 100
0
    link/ether 08:00:27:eb:b0:c0 brd ff:ff:ff:ff:ff:ff
    inet 192.168.23.20/24 brd 192.168.23.255 scope global enp0s8
       valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:feeb:b0c0/64 scope link
       valid_lft forever preferred_lft forever
it23@ubuntu-it23:~$
```

Figure 1.12: Server IP Details (Ubuntu)

```
it23@client01-it23:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:c9:c4:f2 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
       valid_lft 83950sec preferred_lft 83950sec
    inet6 fd17:625c:f037:2:f22a:3af5:7967:faa9/64 scope global temporary dynamic
       valid_lft 86378sec preferred_lft 14378sec
    inet6 fd17:625c:f037:2:a00:27ff:fec9:c4f2/64 scope global dynamic mngtmpaddr
       valid_lft 86378sec preferred_lft 14378sec
    inet6 fe80::a00:27ff:fec9:c4f2/64 scope link
       valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:bc:66:85 brd ff:ff:ff:ff:ff:ff
    inet 192.168.23.11/24 brd 192.168.23.255 scope global noprefixroute enp0s8
       valid_lft forever preferred_lft forever
    inet6 fe80::a85e:d0fe:9c9a:b036/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
it23@client01-it23:~$
```

Figure 1.13: Client 01 IP Details

```
it23@client02-it23:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:ad:41:52 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
       valid_lft 81906sec preferred_lft 81906sec
    inet6 fd17:625c:f037:2:eb15:fef2:e2b8:14d3/64 scope global temporary dynamic
       valid_lft 86309sec preferred_lft 14309sec
    inet6 fd17:625c:f037:2:a00:27ff:fead:4152/64 scope global dynamic mngtmpaddr
       valid_lft 86309sec preferred_lft 14309sec
    inet6 fe80::a00:27ff:fead:4152/64 scope link
       valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:fc:5d:f2 brd ff:ff:ff:ff:ff:ff
    inet 192.168.23.12/24 brd 192.168.23.255 scope global noprefixroute enp0s8
       valid_lft forever preferred_lft forever
    inet6 fe80::5f8c:3e46:562a:f02f/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
it23@client02-it23:~$
```

Figure 1.14: Client 02 IP Details

```
[root@localhost ~]# ping -c 4 192.168.23.11
PING 192.168.23.11 (192.168.23.11) 56(84) bytes of data.
64 bytes from 192.168.23.11: icmp_seq=1 ttl=64 time=0.921 ms
64 bytes from 192.168.23.11: icmp_seq=2 ttl=64 time=12.9 ms
64 bytes from 192.168.23.11: icmp_seq=3 ttl=64 time=2.41 ms
64 bytes from 192.168.23.11: icmp_seq=4 ttl=64 time=2.66 ms

--- 192.168.23.11 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3010ms
rtt min/avg/max/mdev = 0.921/4.748/12.994/4.807 ms
[root@localhost ~]# ping -c 4 192.168.23.12
PING 192.168.23.12 (192.168.23.12) 56(84) bytes of data.
64 bytes from 192.168.23.12: icmp_seq=1 ttl=64 time=0.975 ms
64 bytes from 192.168.23.12: icmp_seq=2 ttl=64 time=13.8 ms
64 bytes from 192.168.23.12: icmp_seq=3 ttl=64 time=1.58 ms
64 bytes from 192.168.23.12: icmp_seq=4 ttl=64 time=1.00 ms

--- 192.168.23.12 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3020ms
rtt min/avg/max/mdev = 0.975/4.357/13.868/5.496 ms
[root@localhost ~]#
```

Figure 1.15: Rocky Server – Test Connectivity

```
it23@ubuntu-it23:~$ ping -c 3 192.168.23.10
PING 192.168.23.10 (192.168.23.10) 56(84) bytes of data.
64 bytes from 192.168.23.10: icmp_seq=1 ttl=64 time=1.10 ms
64 bytes from 192.168.23.10: icmp_seq=2 ttl=64 time=1.64 ms
64 bytes from 192.168.23.10: icmp_seq=3 ttl=64 time=1.15 ms

--- 192.168.23.10 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2093ms
rtt min/avg/max/mdev = 1.102/1.297/1.637/0.240 ms
it23@ubuntu-it23:~$ ping -c 3 192.168.23.100
PING 192.168.23.100 (192.168.23.100) 56(84) bytes of data.
64 bytes from 192.168.23.100: icmp_seq=1 ttl=64 time=0.844 ms
64 bytes from 192.168.23.100: icmp_seq=2 ttl=64 time=1.07 ms
64 bytes from 192.168.23.100: icmp_seq=3 ttl=64 time=1.26 ms

--- 192.168.23.100 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2079ms
rtt min/avg/max/mdev = 0.844/1.055/1.256/0.168 ms
it23@ubuntu-it23:~$ ping -c 3 192.168.23.101
PING 192.168.23.101 (192.168.23.101) 56(84) bytes of data.
64 bytes from 192.168.23.101: icmp_seq=1 ttl=64 time=1.07 ms
64 bytes from 192.168.23.101: icmp_seq=2 ttl=64 time=1.09 ms
64 bytes from 192.168.23.101: icmp_seq=3 ttl=64 time=1.34 ms

--- 192.168.23.101 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2101ms
rtt min/avg/max/mdev = 1.071/1.165/1.337/0.121 ms
it23@ubuntu-it23:~$
```

Figure 1.16: Ubuntu Server - Test Connectivity

```
it23@client01-it23:~$ ping -c 4 192.168.23.10
PING 192.168.23.10 (192.168.23.10) 56(84) bytes of data.
64 bytes from 192.168.23.10: icmp_seq=1 ttl=64 time=2.27 ms
64 bytes from 192.168.23.10: icmp_seq=2 ttl=64 time=1.63 ms
64 bytes from 192.168.23.10: icmp_seq=3 ttl=64 time=1.59 ms
64 bytes from 192.168.23.10: icmp_seq=4 ttl=64 time=0.952 ms

--- 192.168.23.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3337ms
rtt min/avg/max/mdev = 0.952/1.610/2.274/0.467 ms
it23@client01-it23:~$
```

Figure 1.17: Client 01 - Test Connectivity

```
it23@client02-it23:~$ ping -c 4 192.168.23.10
PING 192.168.23.10 (192.168.23.10) 56(84) bytes of data.
64 bytes from 192.168.23.10: icmp_seq=1 ttl=64 time=1.19 ms
64 bytes from 192.168.23.10: icmp_seq=2 ttl=64 time=1.52 ms
64 bytes from 192.168.23.10: icmp_seq=3 ttl=64 time=2.06 ms
64 bytes from 192.168.23.10: icmp_seq=4 ttl=64 time=3.80 ms

--- 192.168.23.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3055ms
rtt min/avg/max/mdev = 1.189/2.141/3.796/1.005 ms
it23@client02-it23:~$
```

Figure 1.18 : Client 02 - Test Connectivity

- These screenshots show the successful ping test between the virtual machines. It confirms that network connectivity has been established correctly.

2. ***DHCP Server - Part B***

2.1 DHCP Server Installation

In this step, the DHCP server package was installed on the CentOS server machine. This service is used to automatically assign IP addresses to client systems within the network.

```
Running transaction
  Installing : boost-system-1.53.0-28.el7.x86_64                                  1/5
  Installing : postgresql-libs-9.2.24-9.el7_9.x86_64                              2/5
  Installing : log4cplus-1.1.3-0.4.rc3.el7.x86_64                                 3/5
  Installing : kea-libs-1.6.0-4.el7.x86_64                                        4/5
  Installing : kea-1.6.0-4.el7.x86_64                                             5/5
  Verifying  : log4cplus-1.1.3-0.4.rc3.el7.x86_64                                 1/5
  Verifying  : kea-libs-1.6.0-4.el7.x86_64                                        2/5
  Verifying  : kea-1.6.0-4.el7.x86_64                                             3/5
  Verifying  : postgresql-libs-9.2.24-9.el7_9.x86_64                              4/5
  Verifying  : boost-system-1.53.0-28.el7.x86_64                                  5/5

Installed:
  kea.x86_64 0:1.6.0-4.el7

Dependency Installed:
  boost-system.x86_64 0:1.53.0-28.el7          kea-libs.x86_64 0:1.6.0-4.el7
  log4cplus.x86_64 0:1.1.3-0.4.rc3.el7         postgresql-libs.x86_64 0:9.2.24-9.el7_9

Complete!
```

Figure 2.1: Installing DHCP (kea)

2.2 DHCP Configuration

The DHCP configuration file was edited to define the network settings. The subnet was configured as 192.168.23.0/24, and the IP address range was set from 192.168.23.100 to 192.168.23.200 as required.

```
{

"Dhcp4": {
    "interfaces-config": {
        "interfaces": ["enp0s8"]
    },

    "lease-database": {
        "type": "memfile",
        "lfc-interval": 3600,
        "name": "/var/lib/kea/kea-leases4.csv"
    },


    "subnet4": [
        {
            "id": 1,
            "subnet": "192.168.23.0/24",

            "pools": [ { "pool": "192.168.23.100 - 192.168.23.200" } ],

            "option-data": [
                {
                    "name": "routers",
                    "data": "192.168.23.10"
                },
                {
                     "name": "domain-name-servers",
                     "data": "8.8.8.8"
                }
            ]
        }
    ]
}
}
~
```

Figure 2.2: DHCP Config file

- This screenshot shows the DHCP server configuration file on the CentOS server. The subnet and IP address range were configured according to the assignment requirement.

```
[root@localhost ~]# kea-dhcp4 -t /etc/kea/kea-dhcp4.conf
2026-04-06 18:59:55.817 INFO  [kea-dhcp4.hosts/2257] HOSTS_BACKENDS_REGISTERED the following host ba
ckend types are available: mysql postgresql
2026-04-06 18:59:55.818 INFO  [kea-dhcp4.dhcpsrv/2257] DHCPSRV_CFGMGR_ADD_IFACE listening on interfa
ce enp0s8
2026-04-06 18:59:55.819 INFO  [kea-dhcp4.dhcpsrv/2257] DHCPSRV_CFGMGR_SOCKET_TYPE_DEFAULT "dhcp-sock
et-type" not specified , using default socket type raw
2026-04-06 18:59:55.820 INFO  [kea-dhcp4.dhcpsrv/2257] DHCPSRV_CFGMGR_NEW_SUBNET4 a new subnet has b
een added to configuration: 192.168.23.0/24 with params: valid-lifetime=7200
[root@localhost ~]# _
```

Figure 2.3: Test the Kea Configuration

```
[root@localhost ~]# sudo systemctl restart kea-dhcp4
[root@localhost ~]# sudo systemctl status kea-dhcp4
● kea-dhcp4.service - Kea DHCPv4 Server
   Loaded: loaded (/usr/lib/systemd/system/kea-dhcp4.service; enabled; vendor preset: disabled)
   Active: active (running) since Mon 2026-04-06 19:00:20 +0530; 3s ago
     Docs: man:kea-dhcp4(8)
 Main PID: 2267 (kea-dhcp4)
   CGroup: /system.slice/kea-dhcp4.service
           └─2267 /usr/sbin/kea-dhcp4 -c /etc/kea/kea-dhcp4.conf

Apr 06 19:00:20 localhost.localdomain kea-dhcp4[2267]: 2026-04-06 19:00:20.867 INFO  [kea-dhcp4...ng
Apr 06 19:00:20 localhost.localdomain kea-dhcp4[2267]: 2026-04-06 19:00:20.868 INFO  [kea-dhcp4...ql
Apr 06 19:00:20 localhost.localdomain kea-dhcp4[2267]: 2026-04-06 19:00:20.868 INFO  [kea-dhcp4...s8
Apr 06 19:00:20 localhost.localdomain kea-dhcp4[2267]: 2026-04-06 19:00:20.868 INFO  [kea-dhcp4...aw
Apr 06 19:00:20 localhost.localdomain kea-dhcp4[2267]: 2026-04-06 19:00:20.869 INFO  [kea-dhcp4...00
Apr 06 19:00:20 localhost.localdomain kea-dhcp4[2267]: 2026-04-06 19:00:20.869 INFO  [kea-dhcp4...ed
Apr 06 19:00:20 localhost.localdomain kea-dhcp4[2267]: 2026-04-06 19:00:20.869 INFO  [kea-dhcp4...=4
Apr 06 19:00:20 localhost.localdomain kea-dhcp4[2267]: 2026-04-06 19:00:20.869 INFO  [kea-dhcp4...sv
Apr 06 19:00:20 localhost.localdomain kea-dhcp4[2267]: 2026-04-06 19:00:20.869 INFO  [kea-dhcp4...ec
Apr 06 19:00:20 localhost.localdomain kea-dhcp4[2267]: 2026-04-06 19:00:20.880 INFO  [kea-dhcp4...ed
Hint: Some lines were ellipsized, use -l to show in full.
[root@localhost ~]#
```

Figure 2.4: Service Running Status

- This screenshot shows that the Ubuntu client machine received an IP address automatically from the DHCP server. The assigned IP address is within the required range.

2.3 DHCP Service Execution and Verification

After completing the configuration, the DHCP service was started and enabled. The Ubuntu client machine was then set to obtain an IP address automatically.

The client successfully received an IP address within the defined range, confirming that the DHCP server is functioning correctly.

```
[root@localhost ~]# firewall-cmd --add-service=dhcp --permanent
success
[root@localhost ~]# firewall-cmd --reload

success
[root@localhost ~]#
```

Figure 2.5: Open Firewall

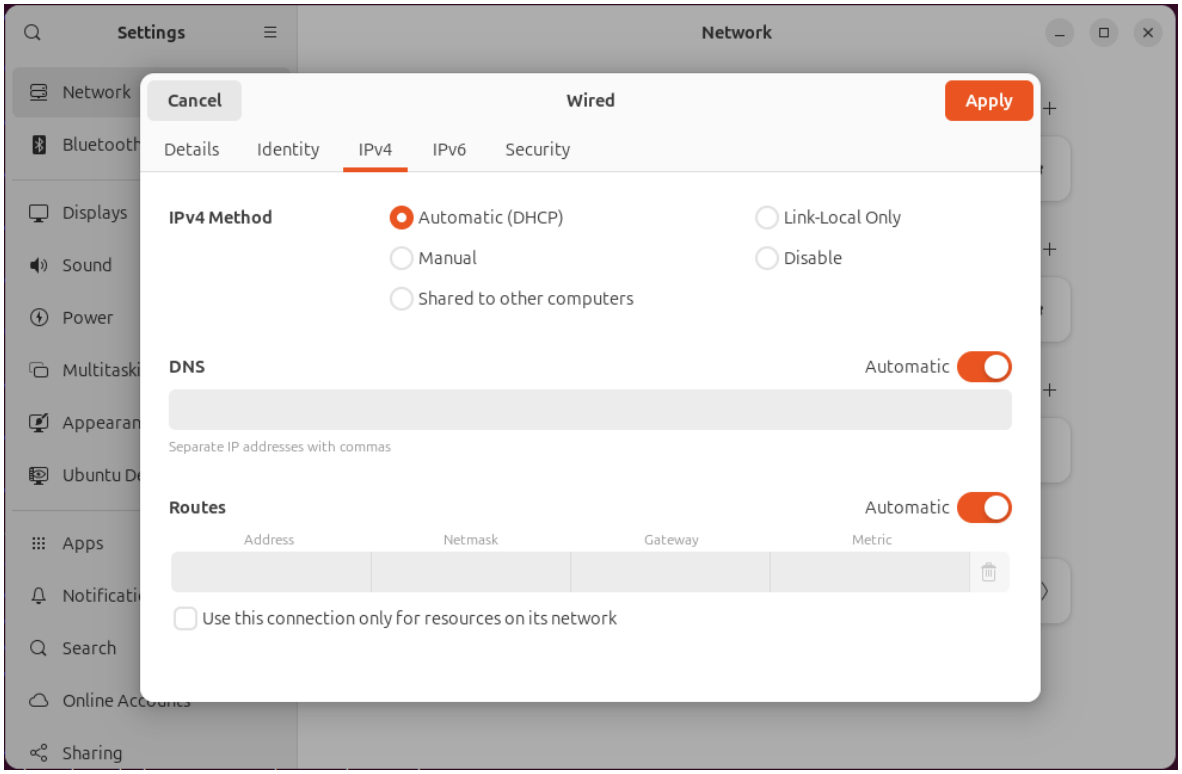

Figure 2.6: Network set to automatic DHCP (In Client 01 & 02)

```
it23@client01-it23:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:c9:c4:f2 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
       valid_lft 83805sec preferred_lft 83805sec
    inet6 fd17:625c:f037:2:3b37:d6cd:bbd8:d3e8/64 scope global temporary dynamic
       valid_lft 86163sec preferred_lft 14163sec
    inet6 fd17:625c:f037:2:a00:27ff:fec9:c4f2/64 scope global dynamic mngtmpaddr
       valid_lft 86163sec preferred_lft 14163sec
    inet6 fe80::a00:27ff:fec9:c4f2/64 scope link
       valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:bc:66:85 brd ff:ff:ff:ff:ff:ff
    inet 192.168.23.11/24 brd 192.168.23.255 scope global noprefixroute enp0s8
       valid_lft forever preferred_lft forever
    inet 192.168.23.100/24 brd 192.168.23.255 scope global secondary dynamic noprefixroute enp0s8
       valid_lft 5159sec preferred_lft 5159sec
    inet6 fe80::a85e:d0fe:9c9a:b036/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
it23@client01-it23:~$
```

Figure 2.7: Client 01 Receiving IP automatically

```
it23@client02-it23:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:ad:41:52 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
       valid_lft 64807sec preferred_lft 64807sec
    inet6 fd17:625c:f037:2:eb15:fef2:e2b8:14d3/64 scope global temporary dynamic
       valid_lft 86255sec preferred_lft 14255sec
    inet6 fd17:625c:f037:2:a00:27ff:fead:4152/64 scope global dynamic mngtmpaddr
       valid_lft 86255sec preferred_lft 14255sec
    inet6 fe80::a00:27ff:fead:4152/64 scope link
       valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:fc:5d:f2 brd ff:ff:ff:ff:ff:ff
    inet 192.168.23.12/24 brd 192.168.23.255 scope global noprefixroute enp0s8
       valid_lft forever preferred_lft forever
    inet 192.168.23.101/24 brd 192.168.23.255 scope global secondary dynamic noprefixroute enp0s8
       valid_lft 6392sec preferred_lft 6392sec
    inet6 fe80::5f8c:3e46:562a:f02f/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
it23@client02-it23:~$
```

Figure 2.8: Client 02 Receiving IP automatically

- These screenshots show that the Ubuntu clients machine received an IP address automatically from the DHCP server. The assigned IP address is within the required range (192.168.23.100 and 192.168.23.101).

3. ***Zimbra Mail Server - Part C***

3.1 Zimbra Installation

In this step, the Zimbra Collaboration Suite was installed on the server machine. Required dependencies were configured prior to installation to ensure proper functionality

```
[root@mail zimbra-install]# wget https://files.zimbra.com/downloads/10.1.0_GA/zcs-NETWORK-10.1.0_GA_
4633.RHEL7_64.20240610085557.tgz
--2026-04-06 20:57:03--  https://files.zimbra.com/downloads/10.1.0_GA/zcs-NETWORK-10.1.0_GA_4633.RHE
L7_64.20240610085557.tgz
Resolving files.zimbra.com (files.zimbra.com)... 18.67.151.23
Connecting to files.zimbra.com (files.zimbra.com)|18.67.151.23|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 446177722 (426M) [application/x-tar]
Saving to: 'zcs-NETWORK-10.1.0_GA_4633.RHEL7_64.20240610085557.tgz'

100%[===================================================>] 446,177,722 18.0MB/s   in 25s

2026-04-06 20:57:30 (17.0 MB/s) - 'zcs-NETWORK-10.1.0_GA_4633.RHEL7_64.20240610085557.tgz' saved [44
6177722/446177722]

[root@mail zimbra-install]#
```

Figure 3.1: Zimbra Installation

```
[zimbra@mail ~]$ zmcontrol status
Host mail.it23.local
        amavis                  Running
        antispam                Running
        antivirus               Running
        convertd                Running
        dnscache                Running
        ldap                    Running
        license-daemon          Running
        logger                  Running
        mailbox                 Running
        memcached               Running
        mta                     Running
        onlyoffice              Running
        opendkim                Running
        proxy                   Running
        service webapp          Running
        snmp                    Running
        spell                   Running
        stats                   Running
        zimbra webapp           Running
        zimbraAdmin webapp      Running
        zimlet webapp           Running
        zmconfigd               Running
[zimbra@mail ~]$
```

Figure 3.2: Verify Zimbra service after installation

- This screenshot shows the Zimbra Collaboration Suite installation on the server. The required mail server components were installed successfully.

3.2 Mail Domain Configuration

A mail domain was created using the format mail.it23.local. Multiple user accounts were created under this domain following a proper naming convention.

These user accounts represent different users within the mail system.

```
[zimbra@mail ~]$ zmprov gad
mail.it23.local
[zimbra@mail ~]$ _
```

Figure 3.3: Mail Domain

- This screenshot shows the created Zimbra mail user accounts using the required naming structure (mail.it23.local).

```
[zimbra@mail ~]$ zmprov -l ca user1@mail.it23.local user1usr
b45965bf-66b7-43c0-9107-29e42b19c74d
[zimbra@mail ~]$ zmprov -l ca user2@mail.it23.local user2usr
1e70b509-94f9-4e9f-bcf0-06823f244b05
[zimbra@mail ~]$
```

Figure 3.4: Create Multiple Users

```
[zimbra@mail ~]$
[zimbra@mail ~]$ zmprov -l gaa
admin@mail.it23.local
spam.b0jws80h@mail.it23.local
ham.zjaemqy_@mail.it23.local
virus-quarantine.bvkevenes6@mail.it23.local
galsync.3q2wmvi77s@mail.it23.local
admin1@mail.it23.local
user1@mail.it23.local
user2@mail.it23.local
[zimbra@mail ~]$
```

Figure 3.5: Checking available Account

- This screenshot shows the created Zimbra mail user accounts using the required naming structure.

3.3 Webmail Interface

The Zimbra web interface was accessed through a browser to test the mail system. Emails were successfully sent and received between users, confirming that the mail server is functioning correctly.

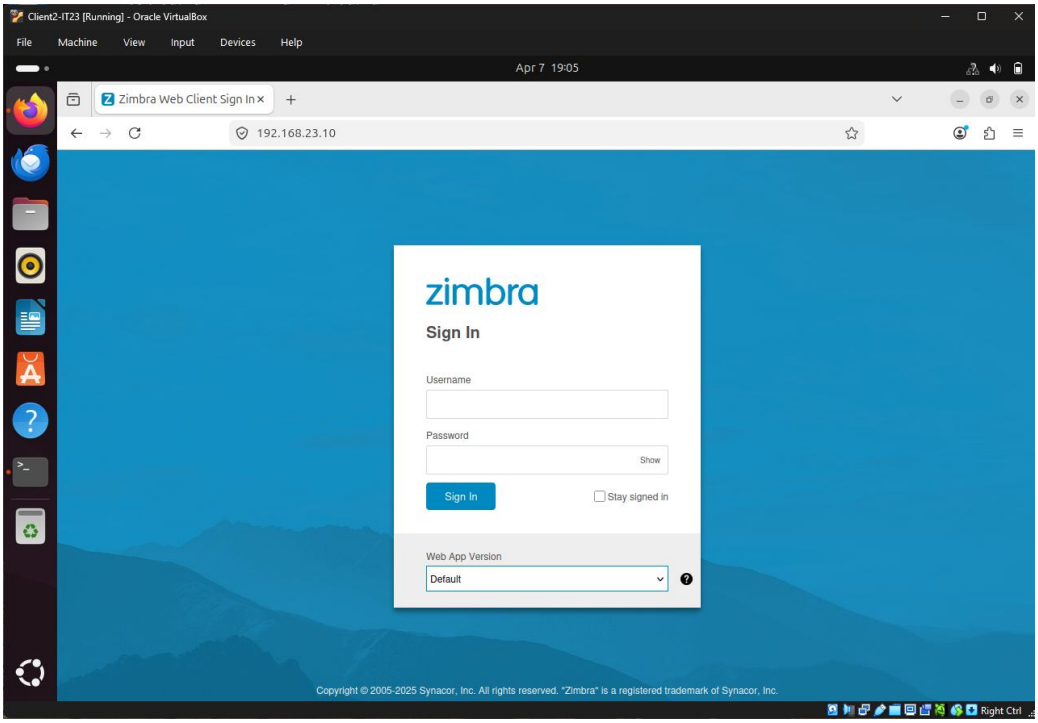

Figure 3.6: Zimbra login page

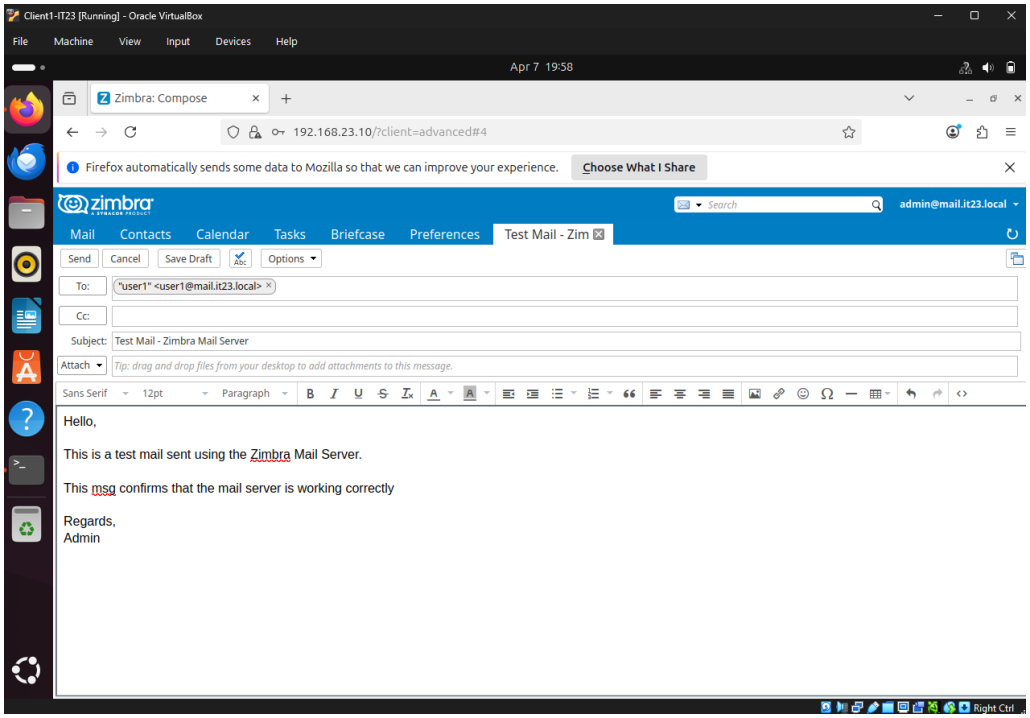

Figure 3.7: Sending mail to User (Client 01 UI)

- This screenshot shows an email sent from Zimbra admin account to another through the Zimbra web interface.

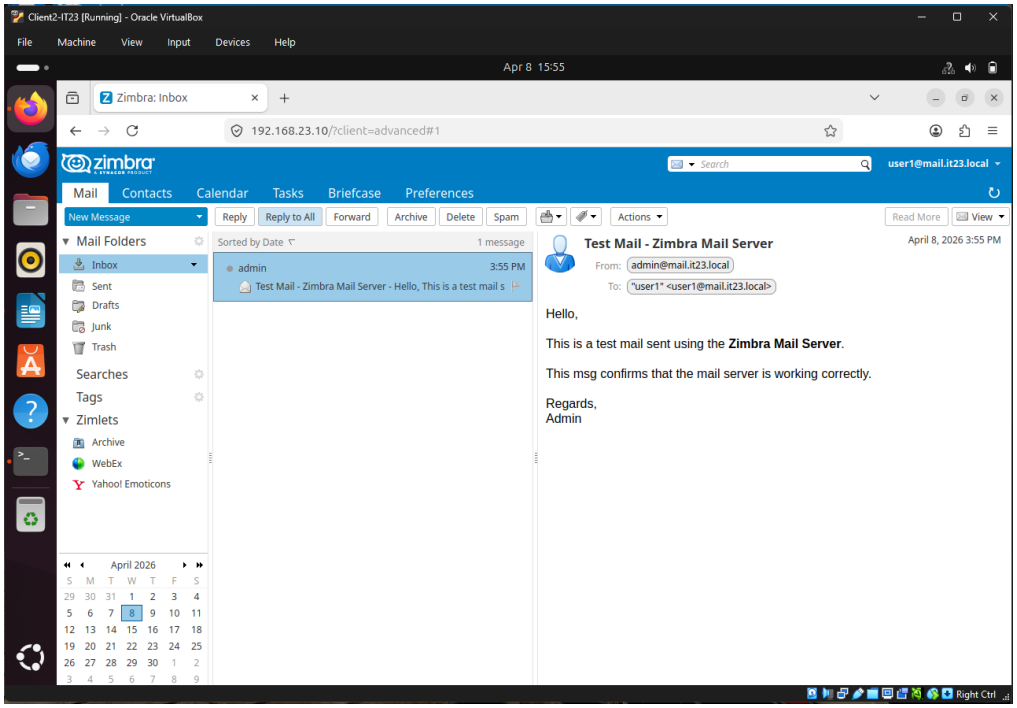

Figure 3.8: Receiving Mail From Admin (Client 02 UI)

- This screenshot shows the successful receipt of the email in another user's inbox, confirming that the mail server is functioning correctly.

4. ***Samba Active Directory Domain Controller - Part D***

4.1 Domain Controller Setup

In this step, Samba was installed and configured as an Active Directory Domain Controller on the server machine. A domain was created using the format it23.local.

```
it23@ubuntu-it23:~$ sudo samba-tool domain provision --use-rfc2307 --interactive
Realm [IT23.LOCAL]:  IT23.LOCAL
Domain [IT23]:  IT23
Server Role (dc, member, standalone) [dc]:  dc
DNS backend (SAMBA_INTERNAL, BIND9_FLATFILE, BIND9_DLZ, NONE) [SAMBA_INTERNAL]:  SAMBA_INTERNAL
DNS forwarder IP address (write 'none' to disable forwarding) [127.0.0.53]:  8.8.8.8
Administrator password:
```

Figure 4.1: Domain provision

- This screenshot shows the configuration of Samba as an Active Directory Domain Controller. The domain was created successfully using the required domain format.

```
● samba-ad-dc.service - Samba AD Daemon
     Loaded: loaded (/lib/systemd/system/samba-ad-dc.service; enabled; vendor preset: enabled)
     Active: active (running) since Mon 2026-04-13 11:08:42 UTC; 44s ago
       Docs: man:samba(8)
             man:samba(7)
             man:smb.conf(5)
   Main PID: 4380 (samba)
     Status: "samba: ready to serve connections..."
      Tasks: 58 (limit: 6965)
     Memory: 196.6M
        CPU: 4.544s
     CGroup: /system.slice/samba-ad-dc.service
             ├─4380 "samba: root process" "" "" "" "" "" "" "" "" "" "" "" "" "" "" "" "" "" "" "" "" "" "" "" >
             ├─4381 "samba: tfork waiter process(4382)" "" "" "" "" "" "" "" "" "" "" "" "" "" "" "" "" "" "" "
```

```
it23@ubuntu-it23:~$ sudo samba-tool ou create "OU=Admins,DC=it23,DC=local"
Added ou "OU=Admins,DC=it23,DC=local"
it23@ubuntu-it23:~$ sudo samba-tool ou create "OU=Staff,DC=it23,DC=local"
Added ou "OU=Staff,DC=it23,DC=local"
it23@ubuntu-it23:~$ sudo samba-tool ou create "OU=Students,DC=it23,DC=local"
Added ou "OU=Students,DC=it23,DC=local"
it23@ubuntu-it23:~$ _
```

Figure 4.3: Create Organizational Units (OU)

- This screenshot shows the created Organizational Units for Admins, Staff, and Students in the domain structure.

```
it23@ubuntu-it23:~$ sudo samba-tool user create admin1 Admin@123
User 'admin1' added successfully
it23@ubuntu-it23:~$ sudo samba-tool user create admin2 Admin@123
User 'admin2' added successfully
it23@ubuntu-it23:~$ sudo samba-tool user create admin3 Admin@123
User 'admin3' added successfully
it23@ubuntu-it23:~$
it23@ubuntu-it23:~$ sudo samba-tool user create staff1 Staff@123
User 'staff1' added successfully
it23@ubuntu-it23:~$ sudo samba-tool user create staff2 Staff@123
User 'staff2' added successfully
it23@ubuntu-it23:~$ sudo samba-tool user create staff3 Staff@123
User 'staff3' added successfully
it23@ubuntu-it23:~$
it23@ubuntu-it23:~$ sudo samba-tool user create student1 Student@123
User 'student1' added successfully
it23@ubuntu-it23:~$ sudo samba-tool user create student2 Student@123
User 'student2' added successfully
it23@ubuntu-it23:~$ sudo samba-tool user create student3 Student@123
User 'student3' added successfully
it23@ubuntu-it23:~$ _
```

Figure 4.4: Create Users

- This screenshot shows the user accounts created for Admins, Staff and Students.

```
it23@ubuntu-it23:~$ sudo samba-tool user move admin1 "OU=Admins,DC=it23,DC=local"
Moved user "admin1" into "OU=Admins,DC=it23,DC=local"
it23@ubuntu-it23:~$ sudo samba-tool user move admin2 "OU=Admins,DC=it23,DC=local"
Moved user "admin2" into "OU=Admins,DC=it23,DC=local"
it23@ubuntu-it23:~$ sudo samba-tool user move admin3 "OU=Admins,DC=it23,DC=local"
Moved user "admin3" into "OU=Admins,DC=it23,DC=local"
it23@ubuntu-it23:~$
it23@ubuntu-it23:~$ sudo samba-tool user move staff1 "OU=Staff,DC=it23,DC=local"
Moved user "staff1" into "OU=Staff,DC=it23,DC=local"
it23@ubuntu-it23:~$ sudo samba-tool user move staff2 "OU=Staff,DC=it23,DC=local"
Moved user "staff2" into "OU=Staff,DC=it23,DC=local"
it23@ubuntu-it23:~$ sudo samba-tool user move staff3 "OU=Staff,DC=it23,DC=local"
Moved user "staff3" into "OU=Staff,DC=it23,DC=local"
it23@ubuntu-it23:~$
it23@ubuntu-it23:~$ sudo samba-tool user move student1 "OU=Students,DC=it23,DC=local"
Moved user "student1" into "OU=Students,DC=it23,DC=local"
it23@ubuntu-it23:~$ sudo samba-tool user move student2 "OU=Students,DC=it23,DC=local"
Moved user "student2" into "OU=Students,DC=it23,DC=local"
it23@ubuntu-it23:~$ sudo samba-tool user move student3 "OU=Students,DC=it23,DC=local"
Moved user "student3" into "OU=Students,DC=it23,DC=local"
it23@ubuntu-it23:~$ _
```

Figure 4.5: Move Users into OUs

- This screenshot shows move user accounts under the required organizational levels in the domain.

5. *Domain Integration and Authentication*

The Ubuntu client machine was successfully joined to the domain. Domain authentication was verified by logging into the client machine using a domain user account.

```
it23@client01-it23:~$ sudo realm join it23.local -U Administrator
Password for Administrator:
it23@client01-it23:~$ realm list
it23.local
  type: kerberos
  realm-name: IT23.LOCAL
  domain-name: it23.local
  configured: kerberos-member
  server-software: active-directory
  client-software: sssd
  required-package: sssd-tools
  required-package: sssd
  required-package: libnss-sss
  required-package: libpam-sss
  required-package: adcli
  required-package: samba-common-bin
  login-formats: %U@it23.local
  login-policy: allow-realm-logins
it23@client01-it23:~$
```

Figure 5.1: Join Domain

- This screenshot shows the Ubuntu client machine being joined successfully and verified join to the domain.

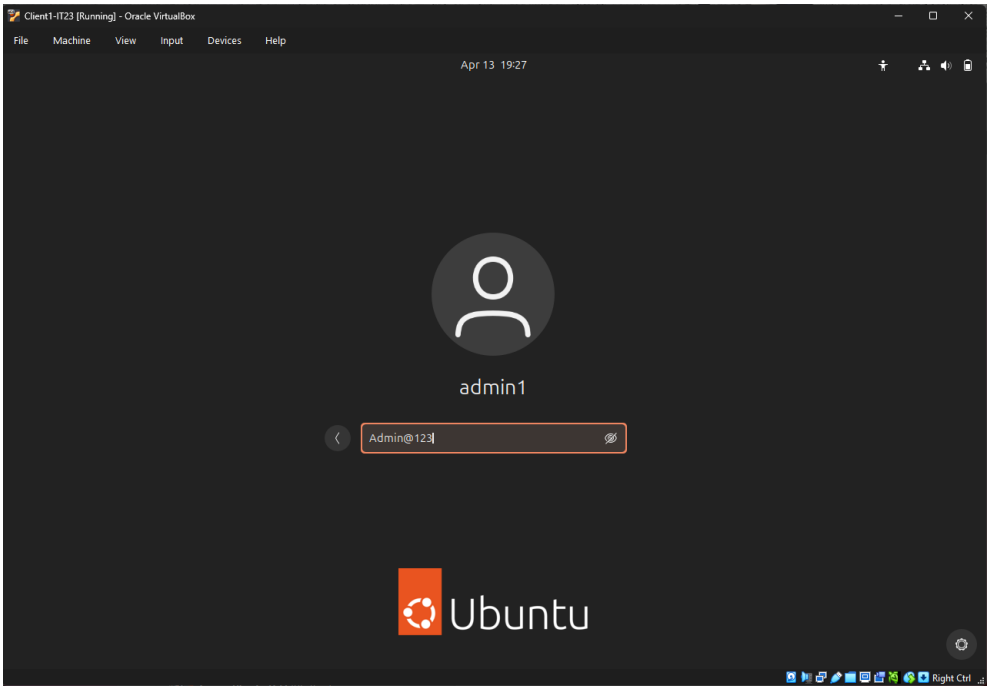

Figure 5.2: Demonstrate domain authentication on Client 01

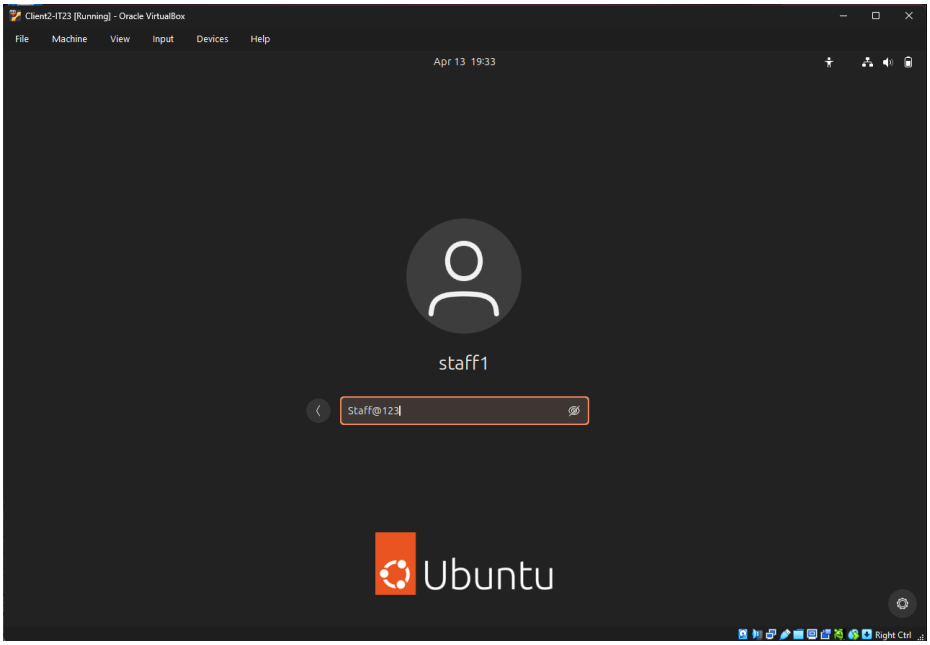

Figure 5.3: Demonstrate domain authentication on Client 02

- These screenshots show successful authentication to the domain using a domain user account from the Ubuntu client machine.

6. ***Web Server – Part E***

6.1 Web Server Installation

In this step, a web server was installed on the Rocky Linux machine. The server was configured to host web content and respond to client requests.

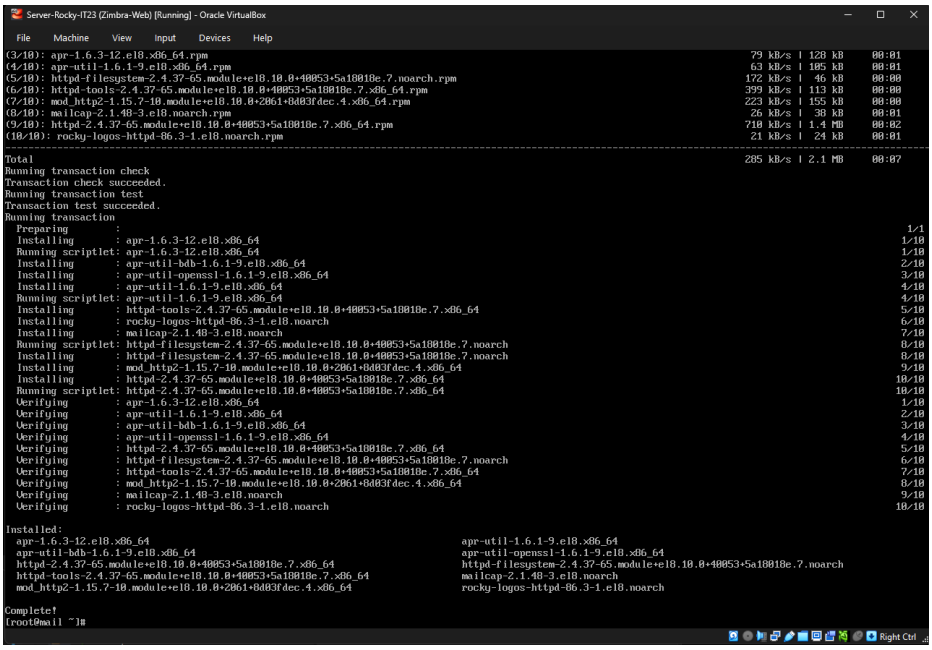

Figure 6.1: Install Apache

```
[root@mail ~]# systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; vendor preset: disabled)
   Active: active (running) since Sun 2026-04-12 21:21:10 +0530; 13s ago
     Docs: man:httpd.service(8)
 Main PID: 214909 (httpd)
   Status: "Running, listening on: port 80"
    Tasks: 213 (limit: 36106)
   Memory: 37.8M
   CGroup: /system.slice/httpd.service
           ├─214909 /usr/sbin/httpd -DFOREGROUND
           ├─214910 /usr/sbin/httpd -DFOREGROUND
           ├─214911 /usr/sbin/httpd -DFOREGROUND
           ├─214912 /usr/sbin/httpd -DFOREGROUND
           └─214913 /usr/sbin/httpd -DFOREGROUND

Apr 12 21:21:10 mail.it23.local systemd[1]: Starting The Apache HTTP Server...
Apr 12 21:21:10 mail.it23.local systemd[1]: Started The Apache HTTP Server.
Apr 12 21:21:10 mail.it23.local httpd[214909]: Server configured, listening on: port 80
[root@mail ~]#
```

Figure 6.2: Verify Service

```
[root@mail ~]# sudo firewall-cmd --permanent --add-service=http
success
[root@mail ~]# sudo firewall-cmd --reload
success
[root@mail ~]# _
```

Figure 6.3: Add HTTP through Firewall

6.2 Webpage Development

A simple webpage was created using HTML and CSS. The webpage displays the text "HELLO NMA" with a blue background, as specified in the assignment requirements.

```
  GNU nano 2.9.8                                        /var/www/html/index.html

<!DOCTYPE html>
<html>
<head>
        <title>Hello NMA</title>
        <style>
                body {
                        background-color: blue;
                        color: white;
                        text-align: center;
                        font-size: 48px;
                        margin-top: 20%;
                        font-family: Arial, sans-serif;
                }
        </style>
</head>
<body>
        HELLO NMA.
</body>
</html>
```

Figure 6.4: Create Web Page

6.3 Web Server Verification

The webpage was accessed from the client machine using a web browser.

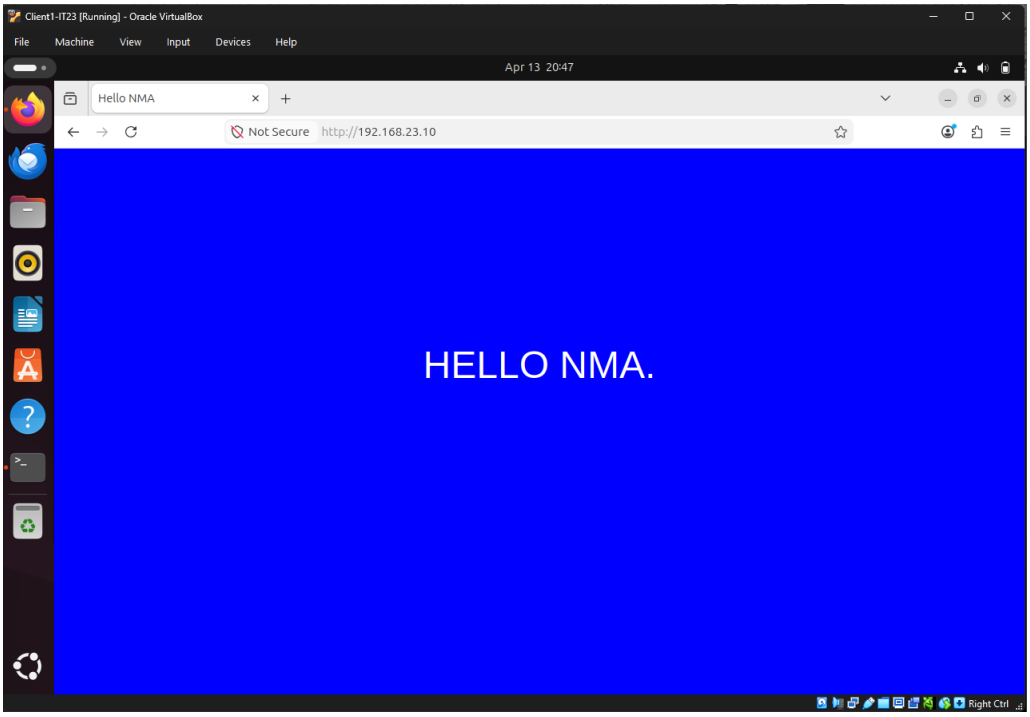

Figure 6.5: Access From Client Browser

- The successful display of the webpage confirms that the web server is functioning correctly.

7. ***Verification***

7.1 DHCP IP Allocation Verification

The DHCP server was verified by confirming that client machines received IP addresses automatically within the configured range.

```
it23@client01-it23:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:c9:c4:f2 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
       valid_lft 76666sec preferred_lft 76666sec
    inet6 fd17:625c:f037:2:afea:9c20:73bb:ebd5/64 scope global temporary dynamic
       valid_lft 86311sec preferred_lft 14311sec
    inet6 fd17:625c:f037:2:a00:27ff:fec9:c4f2/64 scope global dynamic mngtmpaddr
       valid_lft 86311sec preferred_lft 14311sec
    inet6 fe80::a00:27ff:fec9:c4f2/64 scope link
       valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:bc:66:85 brd ff:ff:ff:ff:ff:ff
    inet 192.168.23.11/24 brd 192.168.23.255 scope global noprefixroute enp0s8
       valid_lft forever preferred_lft forever
    inet 192.168.23.100/24 brd 192.168.23.255 scope global secondary dynamic noprefixroute enp0s8
       valid_lft 4666sec preferred_lft 4666sec
    inet6 fe80::a85e:d0fe:9c9a:b036/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
it23@client01-it23:~$
```

Figure 7.1: Client 01 – IP add

```
it23@client02-it23:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:ad:41:52 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
       valid_lft 85914sec preferred_lft 85914sec
    inet6 fd17:625c:f037:2:839b:acda:462f:9df2/64 scope global temporary dynamic
       valid_lft 86329sec preferred_lft 14329sec
    inet6 fd17:625c:f037:2:a00:27ff:fead:4152/64 scope global dynamic mngtmpaddr
       valid_lft 86329sec preferred_lft 14329sec
    inet6 fe80::a00:27ff:fead:4152/64 scope link
       valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:fc:5d:f2 brd ff:ff:ff:ff:ff:ff
    inet 192.168.23.12/24 brd 192.168.23.255 scope global noprefixroute enp0s8
       valid_lft forever preferred_lft forever
    inet 192.168.23.101/24 brd 192.168.23.255 scope global secondary dynamic noprefixroute enp0s8
       valid_lft 7168sec preferred_lft 7168sec
    inet6 fe80::5f8c:3e46:562a:f02f/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
it23@client02-it23:~$
```

Figure 7.2: Client 02 – IP add

- Host-Only interface got IP automatically from DHCP

7.2 Email Communication Verification

The Zimbra mail server was verified by successfully sending and receiving emails between user accounts.

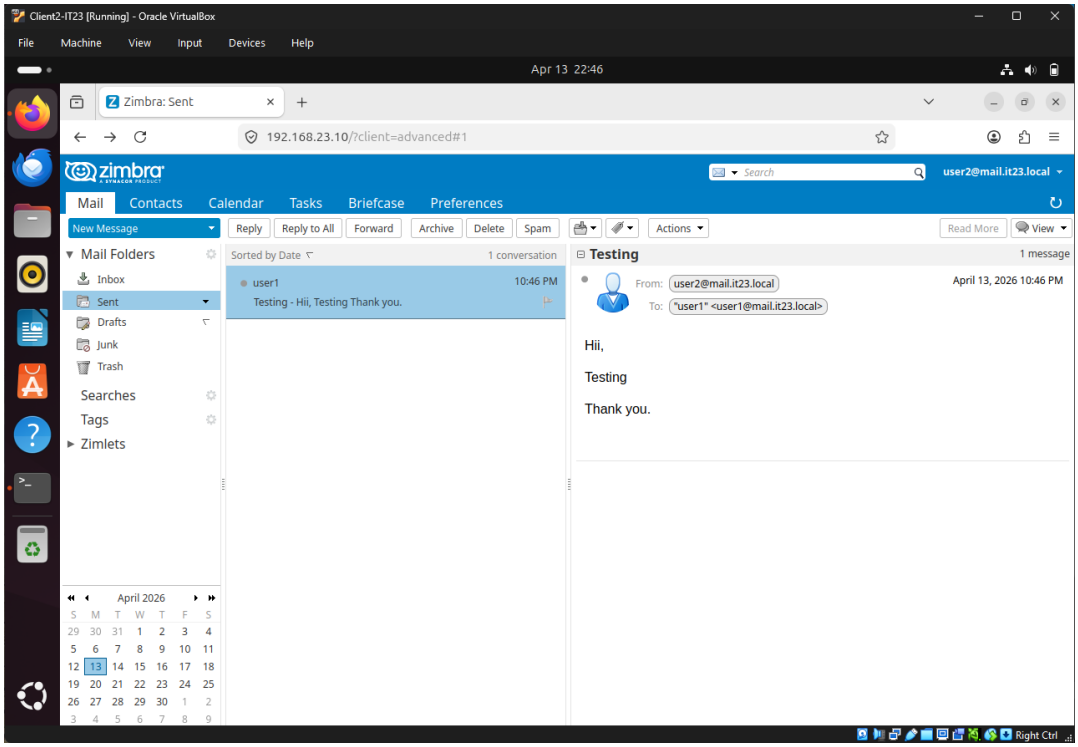

Figure 7.3: Sender Side

- Compose/send email

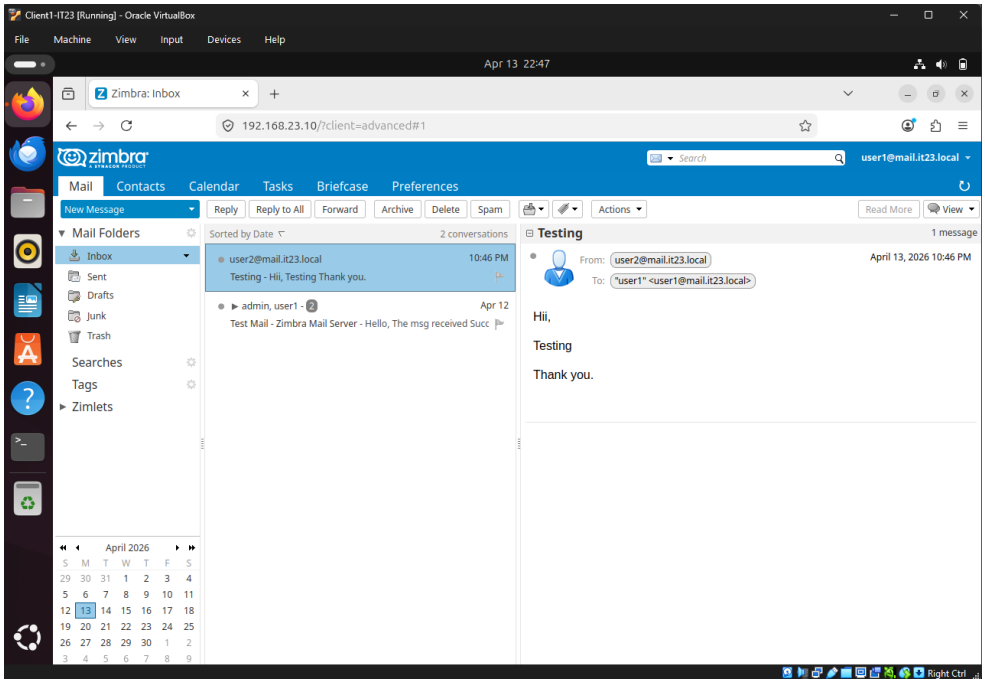

Figure 7.4: Receiver Side

- Inbox showing received email.

7.3 Domain Authentication Verification

The Samba Active Directory Domain Controller was verified by successfully authenticating users and allowing client machines to join the domain.

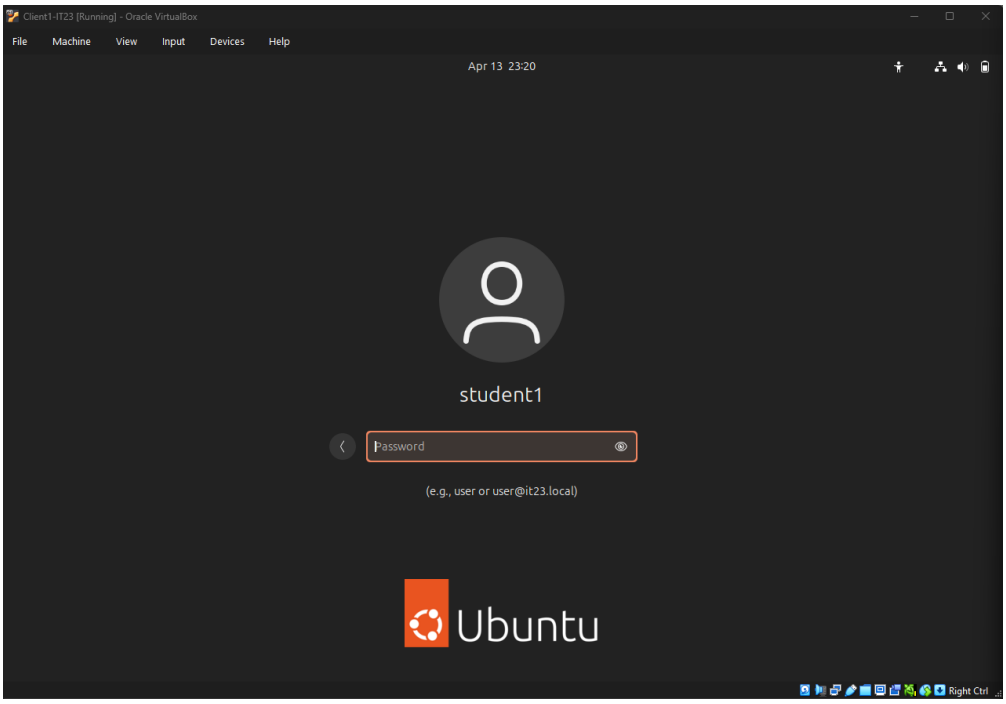

Figure 7.5: Login UI of domain user account

7.4 Web Server Access Verification

The web server was verified by accessing the hosted webpage from the client machine and confirming that it displayed correctly.

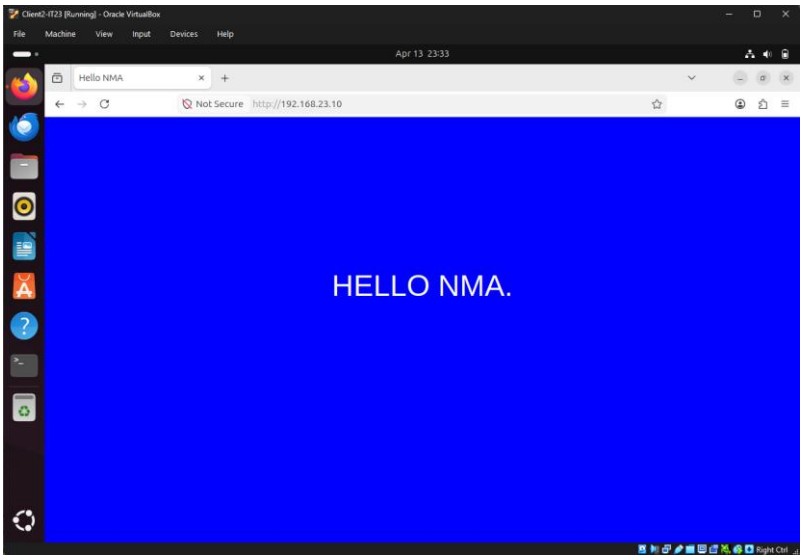

Figure 7.6: Web Page

8. ***Infrastructure Setup Summary***

8.1 Server and Client Environment Overview

Machine Name	Operating System	Version	Role / Services Hosted	IP Address
Server 1	Rocky Linux	8.10	DHCP Server (Kea), Zimbra Mail Server, Web Server (Apache)	192.168.23.10
Server 2	Ubuntu Server	22.04 LTS	Samba Active Directory Domain Controller	192.168.23.20
Client 1	Ubuntu	24.04.3 Desktop	Domain Joined Client / Web & Mail Testing	DHCP Assigned (192.168.23.100)
Client 2	Ubuntu	24.04.3 Desktop	Domain Joined Client / Web & Mail Testing	DHCP Assigned (192.168.23.101)

Table 8.1

8.2 Network Configuration Summary

Network Adapter	Purpose	Configuration
Adapter 1	Internet Access	NAT
Adapter 2	Internal Lab Communication	Host-Only Network

Table 8.2

8.3 Implemented Services Summary

Service	Software Used	Hosted On
DHCP Server	Kea DHCP	Rocky Linux Server
Mail Server	Zimbra Collaboration Suite	Rocky Linux Server
Active Directory / Domain Controller	Samba AD DC	Ubuntu Server
Web Server	Apache HTTP Server	Rocky Linux Server

Table 8.3